

Lichen matted seedbeds inhibit while moss dominated seedbeds facilitate black spruce (*Picea mariana*) seedling regeneration in post-fire boreal forest

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ABSTRACT

The subject of conifer regeneration failure after canopy removing disturbance leading to expansion of ericaceous heaths in boreal forests received considerable attention. However, despite seedbed quality being a key factor in tree regeneration, our understanding of the role of lichen and moss dominated seedbeds on tree seedling regeneration after forest fires remains unclear. Objective of this study was to investigate the effect of time since fire and post-fire cryptogamic (lichen and moss) seedbeds with variable organic matter thickness on black spruce regeneration. We conducted black spruce seeding experiment in black spruce – *Kalmia* forests burned 11, 17 and 37 years ago. In each site we applied three seedbed manipulation treatments (mat-intact, mat-mixed and mat-removed) to test if seedbed manipulation improves spruce regeneration. This experiment was extended to three moss-dominated seedbeds to compare seedbed moisture and temperature effects on spruce regeneration. We also tested potential allelopathic effect of two common lichen and moss species in laboratory bioassay. Seed regeneration was low in all seedbeds. Black spruce germination and seedling growth was lowest in presence of *Cladonia cristatella* followed by *C. stellaris*, *Polytrichum juniperinum* and *Pleurozium schriberi*. HPLC analysis of *C. cristatella* indicated the presence of usnic acid, a common germination inhibiting allelochemical. Moss seedbeds were relatively favourable for spruce regeneration. We conclude that (i) lichen seedbeds inhibit seedling regeneration due to adverse biophysical and chemical (allelopathic) effects, (ii) moss seedbeds facilitate black spruce regeneration by maintaining favourable moisture and temperature, and (iii) seedbed manipulation treatments produce mixed results depending on the dominant cryptogam, organic matter thickness and seedbed moisture and temperature.

1. Introduction

The subject of conifer regeneration failure after canopy removing disturbance such as wildfire and insect infestation leading to expansion of lichen woodlands in the boreal forest received considerable attention (Payette et al., 2000; Payette and Delwaide, 2003; Girard et al., 2008, 2009; Veilleux-Nolin and Payette, 2012). Wildfires modify abiotic and biotic conditions of post-fire seedbeds (Mallik, 2003; Mallik et al., 2010) leaving an ash/charcoal layer and unburned/partially burned organic matter (hereafter called OM) depending on fire severity. Seedling recruitment in severely burned sites is higher than those in low severity burns (Charron and Greene, 2002; Girard et al., 2009). In the absence of high severity natural fires due to aggressive fire suppression, understory ericaceous shrubs, mosses and lichens become dominant in post-fire habitats and interfere conifer regeneration by creating unfavourable seedbeds (Brown and Mikola, 1974; Sedia and Ehrenfeld, 2005, 2006; Fauria et al., 2008; Veilleux-Nolin and Payette, 2012;

Thiffault et al., 2013). This may delay forest succession and in some cases convert conifer forests into ericaceous heaths (Mallik, 1995, 2003; Nilsson and Wardle, 2005; Hyppönen et al., 2013).

Despite seedbed quality being a key factor in seedling regeneration (Siegwart Collier and Mallik, 2010) the role of common boreal seedbeds (lichen and moss-dominated) on conifer regeneration in post fires sites remains unclear (but see Steijlen et al., 1995; Wheeler et al., 2011). Cryptogamic species (lichens and mosses) form a carpet or mat-like structure on top of post-fire seedbeds, which by affecting their physical and chemical properties influence conifer seedling regeneration (Belnap et al., 2001; Steijlen et al., 1995; Sedia and Ehrenfeld, 2003). Mosses can both facilitate and inhibit vascular plant establishment through a variety of mechanisms (Michel et al., 2011; Zackrisson et al., 1997). Moss mats increase soil moisture (Zackrisson et al., 1997), intercept and sequester nutrients from throughfall and litter decomposition (Chapin III et al., 1987), and supply nutrients from senescing tissues to ectomycorrhizal hyphae (Carleton and Read, 1991). Wheeler

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et al. (2011) showed that feather-moss (*Pleurozium schreberi* (Brid.) Mitt.) seedbeds facilitate black spruce seedling regeneration in forest-tundra ecotone. They argued that seedbed composition may influence micro-habitat physical and nutrient condition, and seedling predation. The structure of *P. schreberi* likely protects young seedlings from temperature extremes (Steijlen et al., 1995). Similar results are reported from Swedish boreal forests where feather-mosses add nitrogen to soil in old growth forests (Zackrisson et al., 2004; Lagerström et al., 2007). In contrast, Zackrisson et al. (1997) found that aboveground removal of feather-moss from the seedbed enhances Scots pine (*Pinus sylvestris* L.) germination and growth.

Lichens can also have both positive and negative effects on tree seedling regeneration (During and van Tooren, 1990). Approximately 500 secondary compounds have been reported from lichens, many of which can inhibit conifer regeneration (Mallik, 1987; During and van Tooren, 1990; Steijlen et al., 1995; Zackrisson et al., 1997; Sedia and Ehrenfeld, 2003; Hawkes and Menges, 2003). Both allelopathic effect of lichens and drought or nutrient limitation could suppress germination (Hawkes and Menges, 2003). Growth of planted conifer seedlings can also be inhibited in lichen woodland due to both physical and chemical effects of lichens (Brown and Mikola, 1974). It is critical to understand the causes and consequence of conifer regeneration failure in lichen woodlands, which is rapidly expanding in changing climate (Girard et al., 2008, 2009; Splawinski et al., 2018).

In the boreal forest low-severity fires leave patches of unburned/partially burned OM that may amplify seedbed temperature and moisture fluctuations during summer. The residual OM may contain moss and lichen fragments that allow their rapid recovery after fire (Racine, 1981). Soon after fire, the pioneer crustose lichens and acrocarpous mosses colonize the unburned/partially burned OM followed by foliose and fruticose lichens and pleurocarpous mosses in the later part of succession (Bloom and Mallik, 2004, 2006). Mosses and lichens create unique seedbeds by altering their biophysical and chemical properties. The potential role of lichens and mosses in seedling regeneration may vary depending on residual OM thickness, time since fire and cryptogamic species composition. The mechanism of spruce regeneration failure in post-fire seedbeds is likely multi-faceted involving physical, chemical (allelopathy) and microclimatic conditions of seedbeds. Successful conifer regeneration in lichen-spruce woodlands requires a better understanding of seedbed properties and innovative approach in creating safe sites for tree seedling regeneration (Mallik and Kravchenko, 2018).

Our objective was to investigate the role of lichen and moss dominated seedbeds on black spruce seedling regeneration in post-fire sites by addressing three questions, (i) how does time since fire and seedbed type (lichen vs. moss) and OM thickness affect black spruce regeneration? (ii) does mechanical mixing of cryptogamic mats with OM and removal of lichen or moss mat help black spruce regeneration? and (iii) what role the seedbed physical (moisture and temperature) and chemical (allelopathy) properties play in black spruce seed regeneration? We hypothesised that, (i) seedbed properties will vary depending on time since fire and because cryptogamic mats pose physical barrier for primary roots reaching moist soil, mat-mixing and mat removal would

result better spruce regeneration than mat-intact seedbeds, (ii) moss seedbeds will support higher seedling regeneration than lichen seedbeds by retaining higher soil moisture and lower surface temperature, and (iii) higher phenolic concentration of lichens would lower black spruce germination and seedling growth than mosses.

2. Materials and methods

2.1. Study area

We conducted this study in the greater Terra Nova National Park (TNNP) ecosystem, located in the east-central and north shore eco-regions of Newfoundland, Canada (48° 33' N latitude, 53° 58' W longitude). Climatic conditions are both continental and maritime, due to prevailing westerly winds and close proximity to the Atlantic Ocean. Summers are brief and cool with regional mean summer temperature averaging 12.5 °C and mean annual precipitation ranging from 1000 to 1300 mm (Power, 2000, 2005). TNNP is nearly 80% forested and stands are similar to those of the mainland boreal forests of Canada. Forests are dominated by black spruce and balsam fir (*Abies balsamea* (L.) Mill.) mixed with white birch (*Betula papyrifera* Marsh.), trembling aspen (*Populus tremuloides* Michx.) and, to a lesser extent, larch (*Larix laricina* (du Roi) Koch) and red maple (*Acer rubrum* L.). Black spruce communities are among the most common in the Terra Nova region, occupying 52% of total forest cover. The dominant sub-canopy species associated with black spruce forests and barrens is *Kalmia* and black spruce-*Kalmia* associations occupy nearly 50% of the land base in TNNP (Siegwart Collier and Mallik, 2010). Other ericaceous shrubs such as Labrador tea (*Rhododendron groenlandicum* (Oeder) K.A. Kron and W.S. Judd), *Rhododendron* (*R. canadense* (L.) Torr.) and low-bush blueberry (*Vaccinium angustifolium* Aiton) are also associated with these forest types (Power, 2000). Pre-fire forest floor is composed of lichens (mainly *Cladonia* and *Cladina* spp.), mosses (mainly Acrocarpous and Pleurocarpous), accumulated leaf litter and organic matter (Power, 2000, 2005). High-severity fire has historically facilitated black spruce recovery, producing even-aged stands throughout the region (Power, 2000; Bloom and Mallik, 2006). Effective fire suppression since 1970s decreased stand replacing fires resulting poor conifer regeneration and increased ericaceous heaths (Power, 2000).

2.2. Site selection

We selected three post-fire sites, Rocky Pond (RP), Spracklin Road (SR) and Terra Nova Road (TNR), burned respectively 11, 17 and 37 years ago, to cover varying time since fire, OM thickness and seedbed types (only in SR) to conduct seeding experiments (Table 1). In the early and late post-fire sites, two dominant lichen and moss species were respectively *Cladonia cristatella*, *C. stellaris* and *Polytrichum juniperinum* and *Pleurozium schreberi* (Brid.) Mitt. (Bloom and Mallik, 2006).

Table 1

Location and characteristics of post-fire sites in Rocky Pond (RP), Spracklin Road (SR) and Terra Nova Road (TNR) used for the *in situ* seedbed manipulation experiment in Terra Nova National Park (TNNP), Newfoundland (modified from Mallik and Kravchenko (2016)).

Site	Location	GPS coordinates	Time since fire (yrs)	Burn area (ha)	OM depth (cm) ± se	OM depth range
1	Rocky Pond (RP)	N48°31' 662" W53°58' 967"	11	85	4.0 ± 0.5	0–20
2	Spracklin Road (SR)	N48°31' 953" W54°03' 091"	17	75	6.1 ± 0.7	0–10
3	Terra Nova Road (TNR)	N48°30' 487" W54°07' 143"	37	313	3.9 ± 0.5	0–20

2.3. Field experiments

2.3.1. Effect of time since fire, seedbed type, OM thickness and seedbed manipulation on black spruce regeneration

We followed stratified random design for the *in situ* seeding experiment for both lichen and moss dominated seedbeds. First, only lichen dominated seedbeds were considered in all 3 post fire sites. Due to unavailability of sufficient replication, we considered only one site (SR) for moss-dominated seedbed and analysed data separately.

In this experiment, we considered OM thickness (thin and thick) and seedbed manipulations treatments (intact, mixed and removed) as explanatory variables for both lichen and moss dominated seedbeds following the same procedure. Since OM thickness of a particular microsite was unknown, 12 plots with thin (0–5 cm) and 12 plots with thick (> 5 cm) OM in lichen/moss dominated seedbeds were selected as follows. First, the locations of lichen/moss seedbeds were visually identified and marked following a reconnaissance survey by walking through a 1 ha area. If a microsite met the requirement of thin (0–5 cm) or thick (> 5 cm) OM, then it was selected for determining lichen/moss cover and seedbed manipulation treatment. Lichen/moss cover (%) was estimated visually in 50 × 50 cm quadrats placed in the middle of the survey plot. Four seedbed components (lichen, moss, exposed duff and *Kalmia* litter) were assessed in each 50 × 50 cm plot (Table 2). In the middle of the 50 × 50 plot, we established a 30 × 30 cm treatment plot (exclusively composed of lichen or moss). We measured OM depth from three random locations near the margin of the 50 × 50 cm by extracting the soil cores using a 40 cm long, 5.5 cm diameter Eijkelkamp split tube sampler (Hoskin Scientific, Burlington, Ontario). We measured OM depth from the surface to the most recent charcoal layer in the core. Mean OM thickness of thin OM sites ranged from 1.5 to 4.8 cm and those of thick OM sites ranged from 5.2 to 10 cm with no significant difference across the three age (time since fire) classes (Table 2). We also measured lichen mat thickness at three points from a randomly picked 10 × 10 cm subplot adjacent to the 30 × 30 cm plots (but within the 50 × 50 cm plot). Mat samples collected from these subplots were brought to the laboratory to quantify bulk density.

Three seedbed manipulation treatments (i. mat-intact, control, ii. mat-mixed and iii. mat-removed) were applied randomly in both thin and thick OM dominated with lichen/moss seedbeds. Mat-mixed treatment was applied by mechanically mixing the lichen/moss with OM below. We applied mat-removed treatment by removing the lichen/moss leaving the OM intact. The seedbed manipulation treatments were replicated four times within each OM thickness category (3 treatments × 2 OM thickness × 4 replicates = 24 experimental plots in each site). Only in SR site 24 plots were established in moss seedbeds in addition to the 24 plots of lichen seedbeds.

One hundred black spruce seeds (source: Wooddale Tree Nursery, Grand Falls, NF; 100% viable according to a laboratory viability test) were broadcasted in each plot on July 2, 2014. All the experimental plots were protected from potential herbivore browsing by erecting wire mesh fence. Due to no germination in the first 21 days, all plots were reseeded with an additional 100 seeds on July 24, 2014. The treatment plots were examined for germination three times in the growing season (August 6, 15, and 24, 2014). Percent germination was quantified from the data collected in 2014 and the number of established seedlings was counted on July 1, 2015.

Table 2

Mean ± se of cover (%) of four seedbed components: lichen, moss, exposed duff and *Kalmia* litter at three post-fire study sites (24 samples in each site), in Rocky Pond (RP), Spracklin Road (SR) and Terra Nova Road (TNR), Terra Nova National Park.

Site	Lichen	Moss	Exposed duff	<i>Kalmia</i> litter
RP	79.1 ± 1.9	5.3 ± 2.1	8.4 ± 1.1	7.5 ± 0.8
SR	87.6 ± 1.1	2.9 ± 0.9	3.7 ± 0.6	6.6 ± 0.8
TNR	97.8 ± 0.6	4.2 ± 1.2	0.1 ± 0.7	5.5 ± 0.6

2.3.2. Black spruce regeneration, seedbed moisture and temperature

To determine seedbed moisture and surface temperature in lichen and moss seedbeds, we installed 8 soil moisture and temperature probes (5TM, Decagon scientific, USA) in the buffer area (just outside the 30 × 30 cm plots) in 2 lichen and 2 moss dominated seedbeds in SR site (selected randomly as subsample from lichen and moss plots). In each seedbed type, one probe was placed horizontally at 1 cm and a second probe at 5 cm depth on July 24, 2014 and data acquired at 30 min interval for one month (until August 24, 2014). Five OM samples were collected from each lichen (*Cladonia stellaris*) and moss (*Polytrichum juniperinum*) seedbed (outside the 30 × 30 cm plots) using a steel ring soil sampler (height and diameter 5 cm) to measure bulk density. Bulk density was determined by dividing the volume of the OM with its oven-dried weight.

2.4. Allelopathy of seedbed substrates

We conducted a laboratory bioassay to investigate the potential allelopathic effects of the two common lichen (*Cladonia cristatella* and *C. stellaris*) and moss (*Polytrichum juniperinum* and *Pleurozium schreberi*) species on black spruce germination and primary growth. Ten grams (fresh weight) of each lichen and moss species were macerated separately in a Magic Bullet blender at low speed. Then the macerated samples were placed separately on Petri-dishes (each species considered as an individual treatment) with 10 ml deionized distilled water. Mild maceration was necessary to remove air spaces between live moss and lichen in the Petri-dish in order to enhance contact with black spruce seeds. A layer of Whatman # 1 filter paper was placed above the macerated plant material. A reference treatment was prepared as control by placing a soft cotton layer in a Petri-dish moisten with distilled water covered with a filter paper. Black spruce seeds were surface sterilized with 30% hydrogen peroxide for 10 min followed by washing with distilled water. Ten black spruce seeds were placed on the filter paper. Each treatment was replicated 5 times. To ensure sufficient moisture 5 ml distilled deionized water was added to each dish twice a week. The experiment was maintained at 20 °C with a 12-hour day and night cycle. The plats were checked daily for germination until the end of the experiment (15 d) when root and shoot length of the seedlings were measured. Substrate pH was measured at the end of the bioassay.

2.4.1. Measurement of total phenols

The lichen (*Cladonia cristatella*, and *C. stellaris*) and moss (*Pleurozium schreberi* and *Polytrichum juniperinum*) samples collected from the field were air-dried at room temperature. The samples were ground into a powder with liquid nitrogen in a mortar and pestle followed by sieving through a 500-µm filter. Each 10 g ground lichen and moss sample was mixed with 10 ml HPLC grade methanol and shaken on a rotary shaker at 30 °C for 24 h followed by centrifugation at 14,000 rpm for 10 min. Concentration of total phenols was determined using modified Folin-Ciocalteu method following Inderjit and Mallik (1996). Gallic acid was used as a standard and total phenol was expressed as mg/L gallic acid equivalent. Total phenol concentration was determined using a standard absorbance curve of gallic acid, plotted with the absorbance of 50, 100, 250 and 500 mg/L gallic acid standards at 765 nm. Blanks were produced using deionised distilled water and each sample was measured in triplicate.

2.4.2. HPLC analysis of lichen and moss secondary compounds

Ten grams of each ground and sieved samples of lichen (*Cladonia cristatella* and *Cladonia stellaris*) was mixed with 7 ml HPLC grade methanol and shaken on a rotary shaker at 30 °C for 24 h. The mixture was centrifuged at 14,000 rpm for 10 min in a micro-centrifuge and the extracts collected in glass vials. Three ml distilled water was added to each vial to make the sample methanol: water = 7:3 (total 10 ml). Ten grams of ground lichen was mixed with 7 ml acetone and dichloromethane (50:50) mixture and shaken on a rotary shaker for 24 h

at 30 °C. The solution was filtered through Whatman # 1 filter paper and the solvent was removed using a rotary evaporator at 58 °C. After drying, the residue was dissolved in 7 ml HPLC grade methanol followed by adding 3 ml distilled water. Usnic acid was used as standard because it is a widely cited allelopathic phenolic acid found in lichens. Lichen compounds were analysed using HPLC (Thepnuan et al., 2013). Lichen extracts and standard lichen acids were analysed on a HP 1100 series consisting of a HP G1312A binary pump. Chemical separation was achieved on an Eclipse XDB-C18, 4.6 × 150 mm, and 5 µm column. The solvent A was 1% phosphoric acid in water and solvent B was 100% methanol. The compounds were detected at 210, 256 and 280 nm wavelengths. Identification of compounds was based on retention time.

2.5. Statistical analyses

Differences in lichen cover, mat thickness and bulk density in the three sites were examined by Kruskal-Wallis test followed by Tukey-Kramer-Nemenyi post-hoc test for each variable. The pairwise multiple comparisons of mean ranks for independent samples were carried out using the function “*posthoc.kruskal.nemenyi.test*” in R-package “PMCMR”. We used a two sample *t*-test to identify significant difference in bulk density of OM under lichen and moss mats.

Seed germination and seedling establishment data were analysed in two steps. First, the effects of OM thickness and seedbed manipulation treatments on seed germination and seedling establishment were investigated separately by fitting linear models with seed germination and seedling establishment as dependent variables. Due to unavailability of sufficient replication in each age class analysis was performed separately for 3 post-fire sites (only lichen dominated plots). Because of non-normal residuals of seedling establishment data in SR site, we used Generalized Linear Model (GLM) with Poisson distributed errors. Since, all the plots of a mixed treatment (thick OM) resulted no seedling establishment there were anomalous predicted values in the result. To deal with this, seedling establishment data in SR site were transformed by adding a minute value (0.01) to all observations.

Second, the effects of seedbed type, OM thickness and seedbed manipulation treatments on seed germination and seedling establishment data obtained from moss dominated seedbeds were compared with lichen dominated seedbed (only in SR site). In this case, seed germination was tested using a 3-way ANOVA for a full factorial design. We tested normality of residuals and homogeneity of variances of the groups for germination by Shapiro-Wilk and Levene's tests. Seedling

establishment was tested by fitting a GLM model with Poisson distributed error because residuals were not normally distributed. ANOVA tables for all models used were derived using R function “*anova(model, test = “F”)*” in R library “*car*” (Fox and Weisberg, 2011). Tukey's contrasts was used on seed germination and seedling establishment data to conduct multiple comparisons among treatment groups using *glht* function (general linear hypotheses and multiple comparisons for parametric models, including generalized linear models, linear mixed effects models, and survival model) in R package “*multcomp*” (Bretz et al., 2010; Hothorn et al., 2008).

Seedling survival data were analysed by fitting linear mixed effect model for a repeated measures design using R package lme4 (only in SR site). The model was fitted with number of seedlings in each of four counts as response variables and seedbed type, OM thickness, manipulation treatments and counts (contrasts) as fixed factors. Tukey's contrasts were used on seedling counts (Bretz et al., 2010; Hothorn et al., 2008).

Effects of seedbed type (lichen vs. moss) and OM depth (1 cm vs. 5 cm) on seedbed moisture and temperature (diurnal fluctuations) were investigated by fitting two separate linear mixed effect models for a repeated measures design using R package lme4. For moisture data, the model was fitted with average volumetric moisture content for 30-day period (25 July 2014 to 24 August 2014) as response variable and seedbed type and OM depth as fixed factors. For temperature data, the model was fitted with temperature readings from 12.00 am to 11.30 pm (30 min interval) as a response variable. One randomly selected day (August 4, 2014) from the experimental period was considered to depict diurnal fluctuations of seedbed temperature. For both models, Probe containing plots were considered as random factor in the model to account for the variation in readings each day.

Finally total seed germination data from the laboratory bioassay were analysed by Kruskal-Wallis test because of non-normal residuals. Root and shoot length (cm) were analysed using one-way ANOVA in R package *car*. A two sample *t*-test was used to test significant difference in total phenol among the lichen and moss species.

3. Results

3.1. Lichen cover in the three post-fire sites

Lichen cover was highest in the oldest site (TNR) and lowest in the youngest (RP) site (Fig. 1a). Lichen mat thickness was also significantly

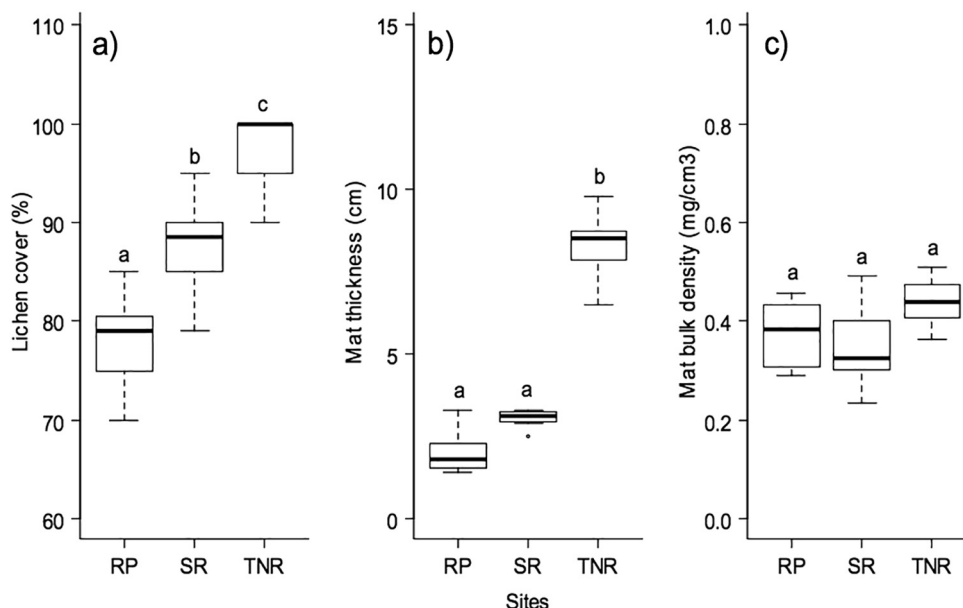


Fig. 1. Boxplots showing (a) lichen cover (%), (b) mat-thickness (cm) and (c) mat bulk density (mg/cm³) in Rocky Pond (RP), Spracklin Road (SR) and Terra Nova Road (TNR) in Terra Nova National Park (TNNP). Different letter(s) above the error bars denote significant difference of mean lichen cover, mat thickness and mat density among them based on Tukey-Kramer-Nemenyi post-hoc test.

higher in the oldest site TNR compared to both RP (t -value = 5.73, $p < 0.01$) and SR site (t -value = 3.236, $p = 0.04$). There was no difference of mat thickness between RP and SR site (Fig. 1b). Lichen mat bulk density did not differ among the three sites (Fig. 1c).

3.2. Black spruce germination and seedling establishment in manipulated lichen seedbeds among sites

3.2.1. Germination

Black spruce germination was low in all sites. Mean ($\pm$ se) germination at RP, SR and TNR were 8.6 ± 1.22 , 10.15 ± 0.88 and $8.54 \pm 1.33\%$ respectively. Seed germination was significantly different among the seedbed manipulation treatments (Fig. 2). There were no significant interactions between OM thickness and seedbed manipulation treatments (Table 3). Highest germination was recorded in mat-intact plots with thick OM in RP and lowest in mat-mixed plots with thin OM. However, the differences were only significant between mat-intact plots with thick OM and mat-mixed plots with thin OM ($t = -3.99$, $p < 0.05$) and between mat-removed plots with thick OM and mat-mixed plots with thin OM ($t = 3.58$, $p < 0.05$; Fig. 2a). In SR site, the highest germination was recorded in mat-mixed plots with thin OM (Fig. 2b). But in TNR site, highest germination was recorded in mat-mixed plots with thick OM with no difference in germination between mat-mixed and mat removed plots with thin OM (Fig. 2c).

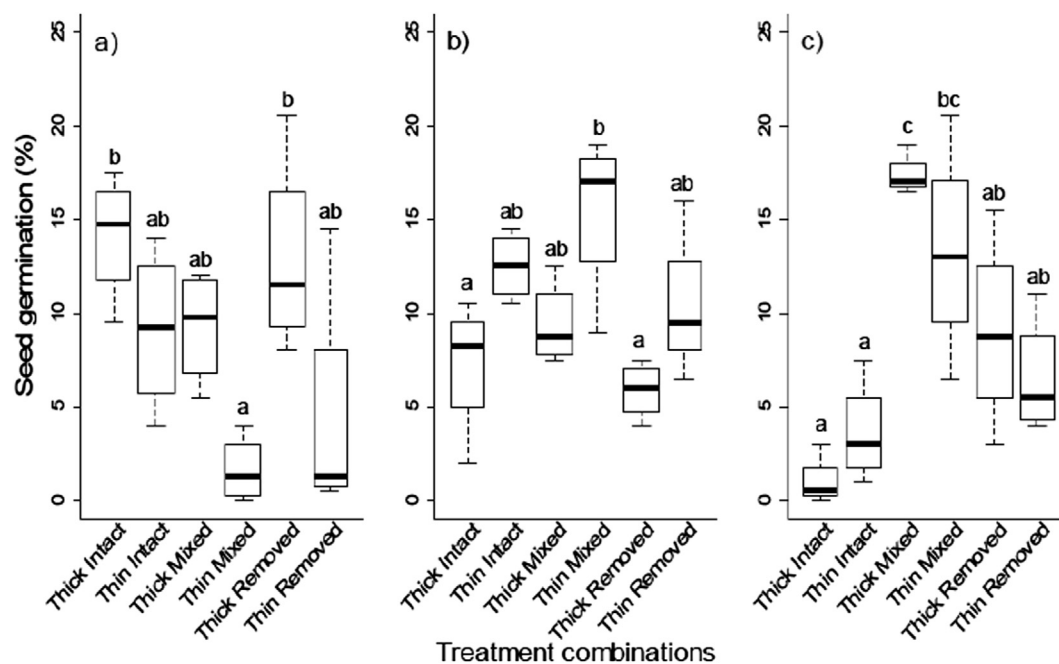


Fig. 2. Boxplots showing black spruce seed germination on mat-intact, mat-mixed and mat-removed treatments in lichen dominated seedbed between thick and thin OM at (a) RP (11 yrs after fire), (b) SR (17 yrs after fire) and (c) TNR (37 yrs after fire). Different letters above the error bars denote significant differences of mean seed germination among treatments at $\alpha \leq 0.05$ based on Tukey's HSD post-hoc test. See Table 3 for summary of statistical test.

3.2.2. Seedling establishment

Seedling establishment was very poor in all three sites (Fig. 3). The effect of OM thickness on seedling establishment was not significant in RP ($F = 3.19$, $p = 0.09$), but it was significant in SR site. Seedbed manipulation treatments had a significant effect on seedling establishment in RP and TNR, but not in SR site (Table 4). In RP, seedling establishment was below 4% in all treatments except mat intact plots with thin OM (slightly above 5%). In SR, seedling establishment was significantly higher in plots with thin OM irrespective of treatments compared to those with thick OM (Fig. 3b). No seedling was established

Table 3

Results of two-way ANOVA comparing black spruce germination in two classes of OM thickness and three seedbed manipulation treatments on lichen dominated seedbeds of three post-fire sites in TNNP, Newfoundland. Significant differences are marked in bold.

Sites	Sources	df	Mean square	F-value	p-value
(1) Rocky Pond	OM thickness	1	297.51	15.14	< 0.01
	Treatment	2	76.59	3.89	0.03
	OM thickness $\times$ Treatment	2	6.64	0.34	0.71
	Error	18	19.65		
(2) Spracklin Road	OM thickness	1	168.01	16.82	< 0.01
	Treatment	2	37.64	3.77	0.04
	OM thickness $\times$ Treatment	2	1.32	0.13	0.88
	Error	18	9.99		
(3) Terra Nova Road	OM thickness	1	10.70	0.8	0.38
	Treatment	2	341.00	25.46	< 0.01
	OM thickness $\times$ Treatment	2	24.80	1.85	0.19
	Error	18	13.40		

in mat-mixed plots with thick OM in this site and very little seedling establishment ($< 1\%$) in mat intact and mat removed plots. Seedbed manipulation treatments responded favourably for seedling establishment in the oldest site (TNP) where highest establishment was recorded

in mat-mixed plots with thick OM followed by lichen mat-removed from thick OM, and mat-mixed on thin OM (Fig. 3c). Here mat-intact plots on thin or thick OM produced the lowest seedling establishment.

3.3. Comparative seedling regeneration in lichen and moss seedbeds in SR site

3.3.1. Germination between lichen and moss seedbeds

Black spruce germination was significantly higher in moss-dominated seedbeds than lichen seedbeds ($F = 8.96$, $p < 0.01$; Table 5).

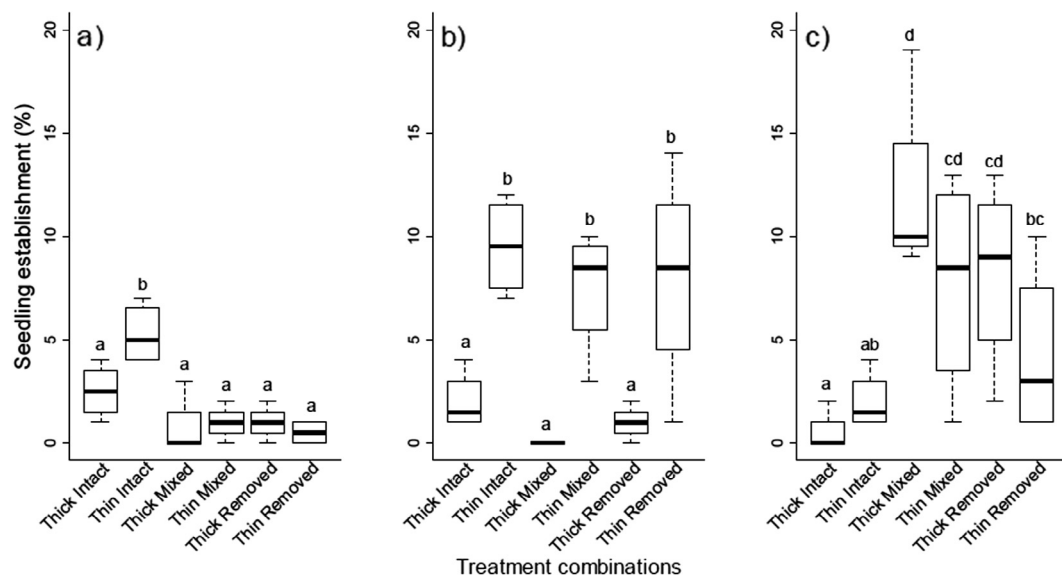


Fig. 3. Boxplots showing seedling establishment on mat-intact, mixed and removed treatments in lichen dominated seedbed between thin and thick OM in (a) RP (11 yrs after fire), (b) SR (17 yrs after fire) and (c) TNR (37 yrs after fire). Different letters above the error bars indicate significant difference of mean seedling establishment (%) at $\alpha \leq 0.05$. Homogeneous subsets are not significantly different based on Tukey's HSD post-hoc test. See Table 4 for summary of statistical test.

Table 4

Results of two-way ANOVA comparing black spruce seedling establishment in two classes of OM thickness and three seedbed manipulation treatments on lichen dominated seedbeds in three post-fire sites in TNNP, Newfoundland. Significant differences are marked in bold.

Sites	Sources	df	Mean square	F-value	p-value
(1) Rocky Pond	OM thickness	1	4.17	3.19	0.09
	Treatment	2	25.04	19.18	< 0.01
	OM thickness $\times$ Treatment	2	5.79	4.44	0.03
	Error	18	1.31		
(2) Spracklin Road	OM thickness	1	75.98	56.04	< 0.01
	Treatment	2	1.13	1.13	0.34
	OM thickness $\times$ Treatment	2	2.10	1.55	0.23
	Error	18	1.36		
(3) Terra Nova Road	OM thickness	1	30.37	1.94	0.18
	Treatment	2	150.04	9.60	< 0.01
	OM thickness $\times$ Treatment	2	21.12	1.35	0.28
	Error	18	15.62		

Table 5

Results of three-way ANOVA comparing black spruce germination by seedbed type, OM thickness and seedbed manipulation treatments. Residuals of the model were normally distributed according to the Shapiro-Wilk test ($W = 0.96$, $p = 0.19$) and variances among the groups were homogeneous according to Bartlett's test ($K\text{-squared} = 11.98$, $p = 0.36$). Significant differences are marked in bold.

Source	Df	Mean square	F	p-value
Seedbed type	1	154.10	8.96	< 0.01
OM thickness	1	0.10	0.01	0.94
Treatment	2	353.95	20.57	< 0.01
Seedbed type $\times$ OM thickness	1	357.50	20.78	< 0.01
Seedbed type $\times$ Treatment	2	149.90	8.71	< 0.01
OM thickness $\times$ Treatment	2	74.95	4.36	0.02
Seedbed type $\times$ OM thickness $\times$ Treatment	2	96.65	5.62	< 0.01
Error	36	17.20		

Germination in lichen and moss seedbeds was 10.1 ± 0.88 and $13.8 \pm 1.86\%$ respectively. Overall, black spruce seed germination did not differ significantly between thin and thick OM ($F = 0.004$, $p = 0.94$). However, there was significant interactions with OM

thickness and seedbed type ($F = 20.78$, $p < 0.01$) and with OM thickness and seedbed manipulation treatments ($F = 4.35$, $p = 0.02$). Germination with thin and thick OM in lichen seedbeds was 12.8 ± 1.14 and $7.5 \pm 0.8\%$ respectively; for moss seedbeds, it was 11.0 ± 1.83 and $16.6 \pm 2.87\%$ respectively.

Seedbed manipulation effects differed when seedbed type (lichen vs. moss) was considered because interaction between manipulation treatments and seedbed type was significant (Table 5). There was no difference in germination among the seedbed manipulation treatments on lichen seedbeds, but mat-mixed treatment with thin OM germination was higher than mat-intact and mat-removed treatments with thick OM (Fig. 4a). In moss seedbed with thick OM, both mat-intact and mat-mixed treatments supported higher germination than the mat-removed treatment (Fig. 4b). Here, the highest germinating was recorded in mat-mixed plots with thick OM and lowest in mat removed plots on thin or thick OM.

3.3.2. Seedling establishment between lichen and moss seedbed

Effects of seedbed type, OM thickness and seedbed manipulation treatments on seedling establishment were significant (Table 6). There was a significant interaction between seedbed type and OM thickness, and seedbed type and manipulation treatments, but not between OM thickness and manipulation treatments (Table 6). In lichen seedbeds, the mat-intact plots on thin OM had the highest seedling establishment where 70.37% of germinated seedlings were established after one year (Fig. 5; Table 6). There was no difference in seedling establishment among treatments of lichen seedbeds with thin OM, but mat-intact and mat-mixed plots with thin OM had significantly higher seedling establishment than those with thick OM. Seedbeds with thin and intact OM had 79% higher seedling establishment than those with thick OM whereas mat-mixed plots with thin OM had 47% seedling establishment. No seedling was established in mat-mixed plots with thick OM (Fig. 5).

On moss seedbeds, the highest seedling establishment was recorded in mat-intact plots with thick OM, where 81% of germinated seedlings established, which is 19% higher than mat-intact plots with thin OM (Fig. 5). Respectively 13 and 41% of germinated seedlings were established in mat-mixed plots with thin and thick OM. The mat-intact plots with thick OM on moss seedbeds had 33 and 87% more seedling establishment than those with thin and thick OM on lichen seedbeds (Fig. 5).

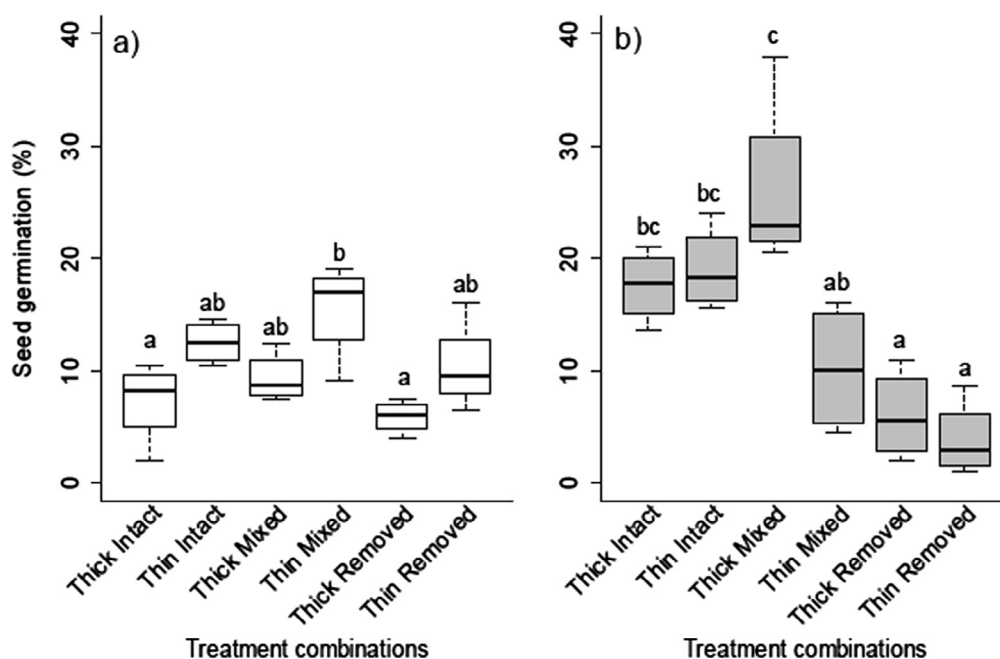


Fig. 4. Boxplots showing black spruce seed germination on mat-intact, mixed and removal treatments between thin and thick OM on (a) lichen and (b) moss dominated seedbeds. Different letters above the error bars denote significant differences at $\alpha \leq 0.05$ determined by Tukey's HSD post-hoc test. See Table 5 for summary of statistical test.

Table 6

Results of three-way ANOVA comparing black spruce seedling establishment by cryptogamic seedbed type, OM thickness and seedbed manipulation treatments. Significant differences are marked in bold.

Source	Df	Mean square	F value	p-value
Seedbed type	1	10.58	6.34	0.01
OM thickness	1	7.08	4.25	0.04
Treatment	2	32.43	19.44	< 0.01
Seedbed type $\times$ OM thickness	1	57.87	34.69	< 0.01
Seedbed type $\times$ Treatment	2	15.24	9.14	< 0.01
OM thickness $\times$ Treatment	2	3.57	2.14	0.13
Seedbed type $\times$ OM thickness $\times$ Treatment	2	10.61	6.36	< 0.01
Error	35	1.67		

3.3.3. Seasonal and post-winter seedling survival in lichen and moss seedbeds

The number of seedlings on lichen seedbeds was consistently lower than on moss seedbeds in both first and second summer except the mat-mixed treatment (Fig. 6; Table A.1). There was an increase in the number of seedlings from first count to third count (end of the first growing season) in all treatments (mat-intact $z = 5.31$, $p < 0.01$, mat-mixed: $z = 7.93$, $p < 0.01$, mat-removed: $z = 5.11$, $p < 0.01$). The number of overwintering seedlings declined in all treatments of lichen seedbeds, but the difference was significant only between the third (pre-winter) and final count (post-winter) in mat-mixed plots, meaning that overwinter mortality was comparatively higher in mat-mixed treatments on lichen seedbeds (mat-intact: $z = -2.7$, $p = 0.72$, mat-mixed: $z = -5.76$, $p < 0.01$, mat-removed: $z = -2.49$, $p = 0.62$; Fig. 6a).

The lowest germination and seedling establishment occurred in mat-

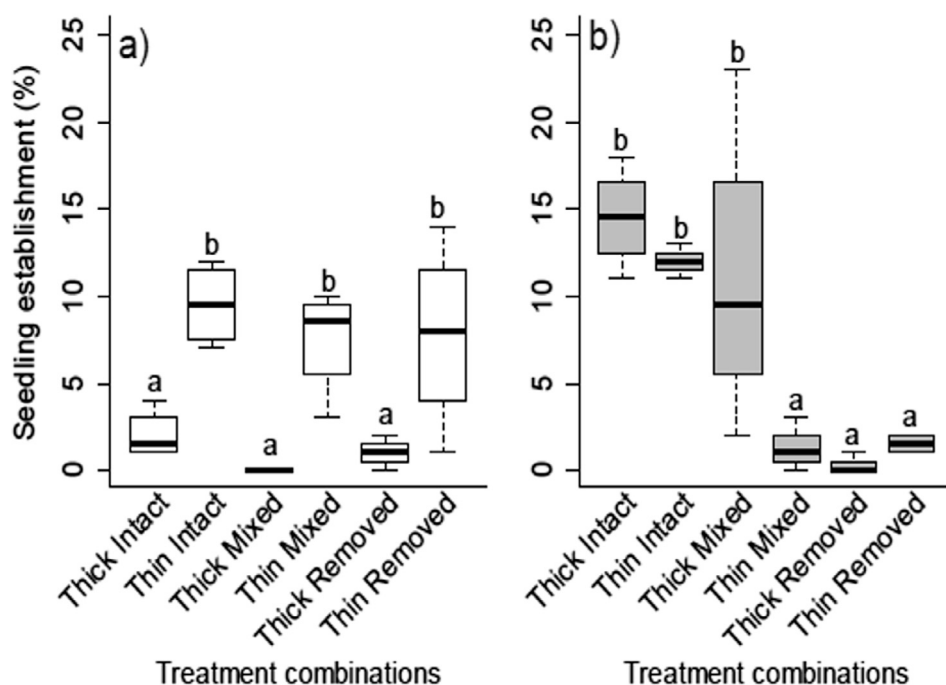


Fig. 5. Boxplots showing seedling establishment on mat-intact, mixed and removed treatments between thin and thick OM on (a) lichen and (b) moss seedbeds. Different letter(s) above the error bars denote significant difference at $\alpha \leq 0.05$. Homogeneous subsets are not significantly different based on Tukey's HSD post-hoc test. See Table 6 for summary of statistical test.

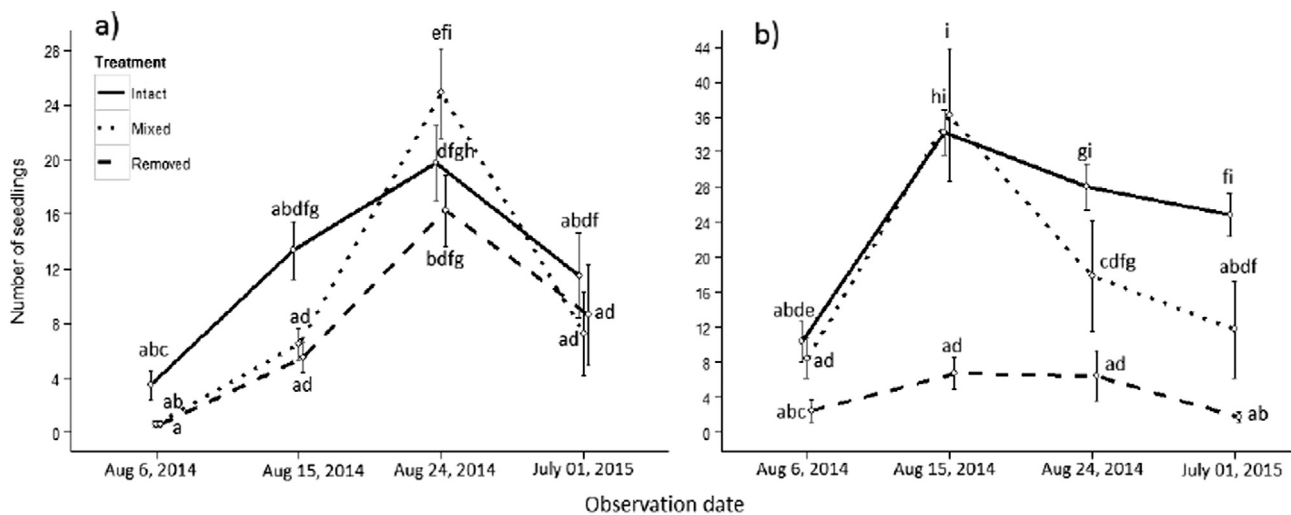


Fig. 6. Mean ($\pm$ se) number of survived seedlings among the seedbed manipulation treatments in (a) lichen and (b) moss seedbeds across four repeated counts in Spracklin Road site. Different letter (s) above the error bars denote significant difference at $\alpha = 0.05$. Homogenous subsets are not significantly different based on Tukey's HSD multiple comparison on number of seedlings by 3-way interaction of seedbed type, treatments and contrasts. See Appendix Table A1 for summary of statistical test.

removed plots on moss dominated seedbed and it did not differ across the counts (Fig. 6b). There were significant differences in the number of winter-survived seedlings between mat-intact and mat-removed ($z = -4.331$, $p = 0.001$) but no difference between mat-intact and mat-mixed ($z = -2.45$, $p = 0.32$) and between mat-mixed and mat-removed plots ($z = -1.87$, $p = 0.73$) of moss seedbeds (Fig. 6b). The number of seedlings increased significantly from the first count to second count in mat-intact ($z = 7.81$, $p < 0.01$) and mat-mixed ($z = 9.08$, $p < 0.01$) plots. However, there was a significant decline in the number of seedlings from the second count to third count (end of the first growing season) in mat-mixed plots ($z = -6.01$, $p < 0.01$). There was no difference in the number of seedlings survived between third count (pre-winter) to fourth count (post-winter) in all treatments (mat-intact: $z = -1.02$, $p = 1.0$; mat-mixed: $z = -2.0$, $p = 0.92$; mat-removed: $z = -1.50$, $p = 0.99$) (Fig. 6b). In other words, overwinter seedling survival rates were similar regardless of treatments in moss-dominated seedbeds.

3.3.4. Bulk density of OM under lichen and moss seedbeds

Bulk density of OM collected under moss mats was significantly higher than that under lichen mats ($t = -3.9901$, $p = 0.016$). Mean ($\pm$ se) bulk density of OM from lichen and moss seedbeds were 0.82 ± 0.06 and 2.09 ± 0.32 mg/cm³ respectively.

3.4. Seedbed moisture

Seedbed moisture was different among seedbed types (lichen vs. moss) and between OM depths (Fig. 7). Soil moisture was higher in moss seedbed than lichen seedbed at 1 cm depth. In lichen seedbed, it was consistently below 10% (range 5–10%) even on rainy days (Aug 10–15 and Aug 20–24, 2014) (Fig. 7a). In moss seedbeds however, even in dry periods, soil moisture was consistently above 10%, mostly between 15 and 25% (Aug 3–8 and Aug 16–18, 2014) (Fig. 7b). Rainfall in July and August 2014 in TNNP area show periodic droughts that coincides with low seedbed moisture at 1 cm depth and surface temperature (Fig. A.1). Soil water contents at 5 cm depth in lichen and moss dominated seedbeds were fairly similar in the growing season (Fig. 7a, b).

3.5. Seedbed temperature

Lichen seedbeds experienced a greater fluctuation of diurnal

temperature than moss seedbeds specially during the drought period (Fig. 8). At 1 cm depth diurnal temperature ranges in lichen and moss seedbeds were respectively 35–40 °C and 20 to 25 °C. There was little diurnal temperature fluctuation at 5 cm depth in lichen and moss seedbeds, which was substantially lower than 1 cm depth (Fig. 8a, b).

3.6. Laboratory bioassay

3.6.1. Seed germination on lichen and moss substrates

Petri-dishes containing the lichen *Cladonia cristatella* had significantly lower germination (mean $\pm$ se, $7.60 \pm 0.40\%$) than all other treatments including control. Germination in both *P. juniperinum* and *P. schreberi* was 10 ± 0.0 and that in lichen *C. stellaris* was 9.4 ± 0.68 with no significant difference among them.

3.6.2. Seedling root and shoot length on lichen and moss substrates

Root length of black spruce seedlings germinated in lichen *C. cristatella* was significantly lower than in other treatments including control (Fig. 9). Shoot length of black spruce seedlings grown with lichen *C. cristatella* was also significantly lower than in control and in moss treatments (Fig. 9).

3.7. Lichen and moss-mat pH and total phenol

The Petri-dishes treated with *Cladonia cristatella* were more acidic (mean pH 4.00 ± 0.06) than those containing *C. stellaris* (pH 4.03 ± 0.09), *P. schreberi* (pH 4.10 ± 0.06) and *P. juniperinum* (pH 5.07 ± 0.03). Total phenol (TP) content varied from 81 to 187 mg/L among the cryptogam species but *C. cristatella* had the highest TP (187 mg/L GAE) and *P. schreberi* had the lowest TP (81 mg/L).

3.8. HPLC analysis for lichen and moss secondary compounds

Presence of usnic acid from selected lichen samples were identified based on retention time for major peaks (Table A.3). Usnic acid standard solution produced a peak at 22.623 min (area 1424.99 mAU*s and peak height 244.69 mAU). Methanol extraction of *C. cristatella* produced two major peaks at 22.501 and 23.082 min and the acetone-dichloromethane extraction produced one additional major peak at 3.814 min (area = 1793.71 mAU*s, height = 323.135 mAU). Since usnic acid produced peak at 22.623 min, which is close to the first major peak of *C. cristatella*, the presence of usnic acid was assumed in

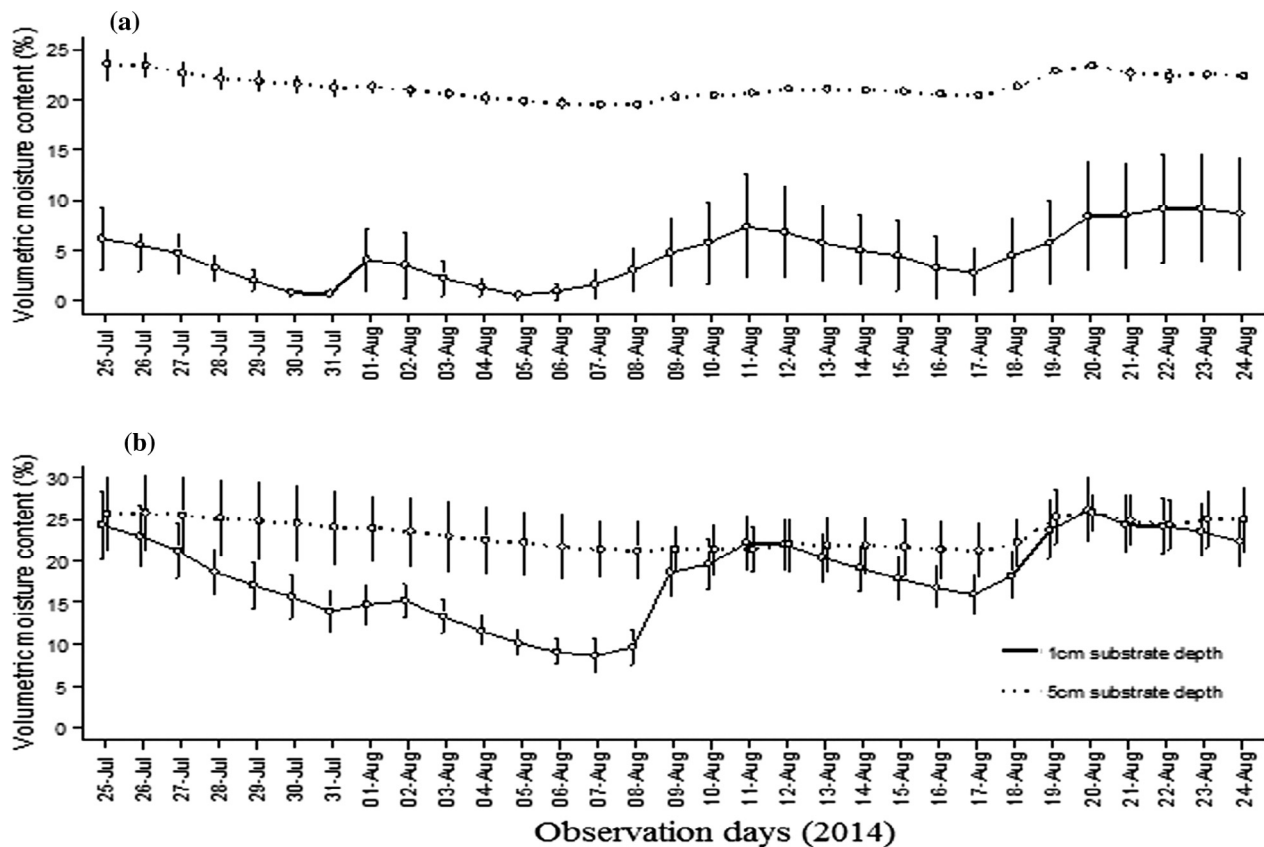


Fig. 7. Volumetric moisture content (%) at 1 cm and 5 cm depths in (a) lichen dominated and (b) moss dominated seedbeds. Data points represent the mean of two readings $\pm$ se.

this lichen (Fig. A.2). Similarly, methanol extraction of *C. stellaris* produced a peak at 22.505 min (area = 2902.61 mAU*s, height = 517.19). Usnic acid produced peak at the same retention time and its presence in *C. stellaris* was suggested, peak area and height of *C. stellaris* was higher than that of *C. cristatella* (Table A.3; Fig. A.2).

4. Discussion

4.1. Time since fire, lichen mat thickness and spruce regeneration

We found increased lichen cover and lichen mat thickness with time since fire and black spruce germination and seedling establishment negatively related to mat thickness providing support of our first hypothesis that seedbed properties vary depending on time since fire. Increased accumulation of organic matter in lichen seedbeds creates unfavourable seedbeds for spruce regeneration (see further discussion comparing lichen vs moss seedbed quality in terms of seedbed moisture and temperature). Irrespective of time since fire (i.e., age) seedling regeneration was poorer on thick than thin lichen mat. Although the overall seedling regeneration was lowest in the oldest site (TNR), thin lichen mat in intermediate age site (SR) had higher seed regeneration than that of the youngest site (RP). We suspect that this difference in seedling regeneration is more to do with the thickness of undecomposed and partially decomposed organic matter below the lichen mat that controls seedbed quality by moisture-temperature change and allelopathy, which is a function fire severity rather than time since fire (Bloom and Mallik, 2006; Siegwart Collier and Mallik, 2010; Mallik and Kravchenko, 2018). While we cannot completely exclude the possibility that the thick OM under lichen might be from unburned/partially burned forest floor litter consisting of a mixture of *Kalmia* litter, mosses and lichen it is more likely that because of slow decomposition of lichens due to low soil moisture and high temperature the accumulated

OM under lichen mat is of lichen origin. Although conifer regeneration on moss and lichen dominated seedbed has been reported earlier (Steijlen et al., 1995; Wheeler et al., 2011; Stuijver et al., 2014) this is the first study that separates the effect of lichen mats from OM underneath on spruce regeneration. It is clear that the biophysical and chemical nature of the surface material (lichen and moss mats) on which the seeds first lodge, strongly affect spruce germination and seedling establishment.

4.2. Seedbed manipulation effects on germination and seedling establishment

We hypothesized that if lichen mat pose physical barrier to black spruce germination and seedling establishment, then mixing the mat with OM or removing it completely would result in higher germination and seedling establishment. Mat intact microsites provided favourable germination and establishment conditions while mat manipulation treatments (mat mixing and mat removal) eliminated the favourable microsite condition i.e., favourable seedbed moisture and temperature. Several authors suggested that surface cracks and fissures on unburned lichen mats provide favourable conditions for seedling establishment (Sirois, 1995; Power, 2005). A possible reason for successful seedling establishment in dry lichen mats might be that cracks in lichen mats provide safe sites (*sensu* Harper, 1977) for seeds and seedlings with favourable soil moisture and temperature with reduced distance to mineral soil. Germinated seedlings within cracks have a greater chance of reaching mineral soil than those on the surface. Mechanical mixing of lichen mat with OM and removal of lichen mat in early and mid-post-fire sites eliminated the cracks and fissures thereby resulting in lower seed germination. Kayes (2008) also reported lower germination in manipulated plots in early post-fire sites in SW Oregon, USA. Studying in the same area as this study Siegwart Collier and Mallik (2010) found

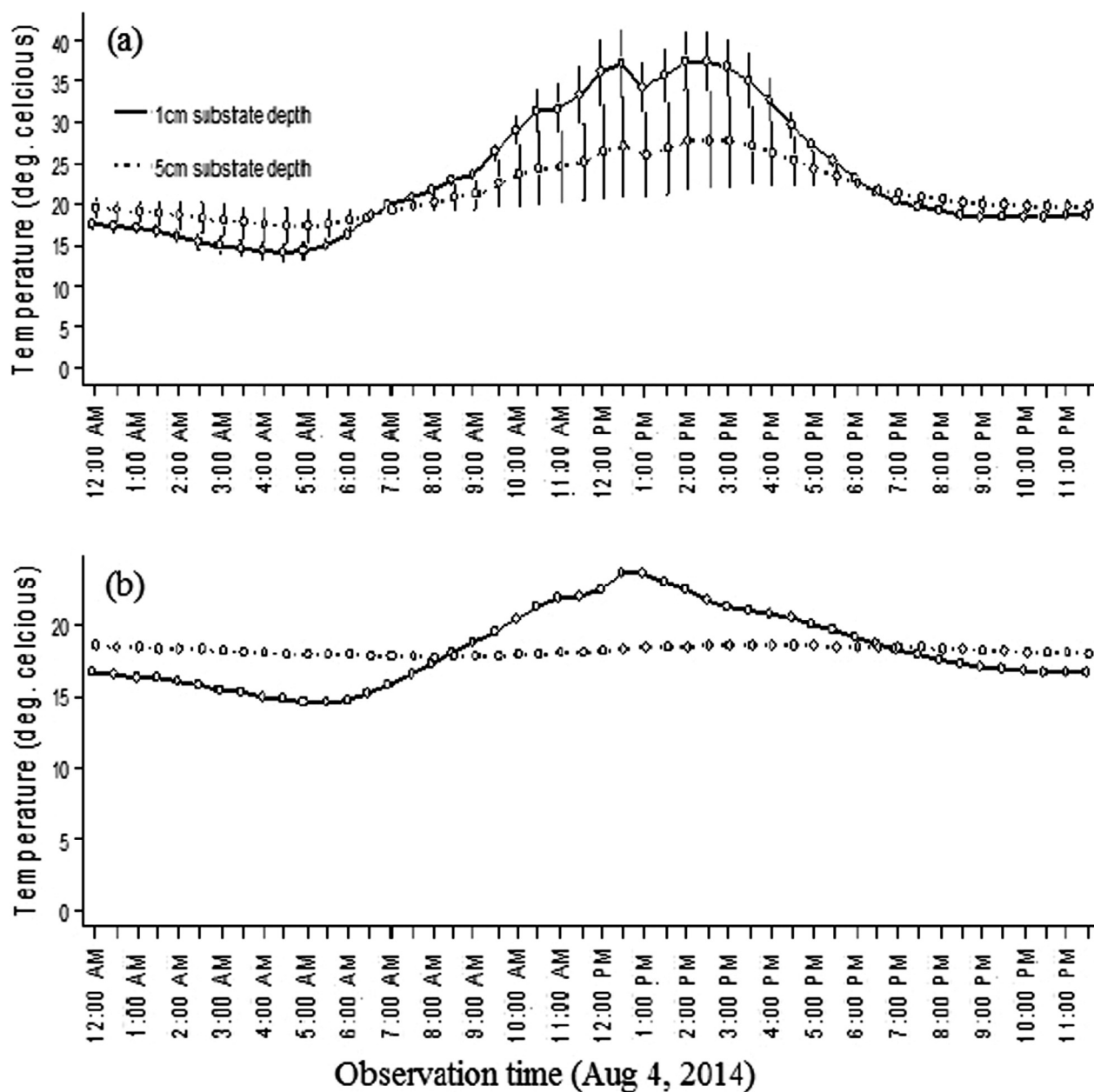


Fig. 8. Diurnal temperature (°C) fluctuations between 1 cm and 5 cm depths in (a) lichen and (b) moss dominated seedbeds. Data points represent the mean of two readings $\pm$ se.

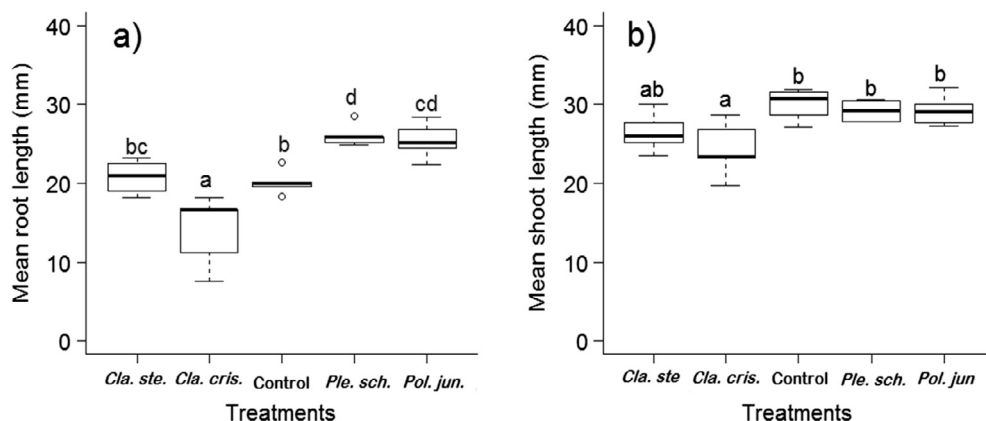


Fig. 9. Boxplot showing difference of (a) root length (mm) and (b) shoot length (mm) of seedling growing on selected cryptogamic species in laboratory bioassay where *Cla. ste.*, *Cla. cris.*, *Ple. sch.* and *Pol. jun.* denote *Cladina stellaris*, *C. cristatella*, *Pleurozium schreberi* and *Polytrichum juniperinum* respectively. Different letter(s) above the error bars denote significant difference at $\alpha \leq 0.05$ level of Tukey's HSD post-hoc test.

that thin residual OM (≤ 2 cm) had more spruce seedlings than those with thick OM.

Contrary to the results of early and mid-post-fire sites (RP and SR), seedling establishment was higher in mat mixed plots in the late post-fire site (TNR). This difference can be explained by the difference in lichen mat thickness. Lichen mat thickness in the late post-fire site was higher than that of early and mid-post-fire sites. Thick lichen mats of intact plots might have impeded seed contact with mineral soil, whereas mat-mixed plots might have eliminated this physical barrier and improved germination. The mat-removed treatment eliminated ground cover, rendering seeds exposed to direct sunlight, higher temperature and desiccation from seedbed moisture loss. Removal of ground cover also exposed the seedlings to potential herbivore damage. Mallik and Kravchenko (2016) reported higher herbivore damage of planted spruce seedlings in scarified and micro-site mulching (mixing OM with mineral soil) plots than undisturbed control plots. Extreme surface temperature due to exposed OM surface in post-fire sites was also suggested by Johnstone and Kasischke (2005). They argued that the black charred OM adsorb and retain heat from the sunlight causing excessive temperature rise and moisture loss. This might be the case in mat removed and mat mixed plots.

In this study, most germination occurred during the rainy period of August 15 to 24, 2014. However, during this time in late post-fire site (TNR), we found no difference in seedling establishment between thin and thick OM seedbed. Germination and seedling establishment were significantly lower in mat intact plots than mat mixed plots regardless of OM depth. Removal of the physical barrier of the thick lichen mat in this site might have enhanced germination and seedling establishment. Several authors reported improved growth of planted black spruce in artificially created safe sites by mixing OM with part of mineral soil (Walker and Mallik, 2009; Mallik and Kravchenko, 2016).

4.3. Moss facilitation

Seed germination and seedling establishment was higher in moss seedbed than lichen seedbed. Wheeler et al. (2011) reported similar results in pleurocarpous moss seedbed compared to bare soil and *Cladonia* lichen seedbed in forest-tundra ecotone. In this study the early post-fire site (RP) dominant mosses were mostly acrocarpous including *Polytrichum juniperinum* and *P. commune* and dominant lichens were foliose and fruticose morphotypes including different species of *Cladonia* and *Cladina*. Our data suggests that moss facilitation was mostly mediated by high soil moisture and low surface temperature. Wheeler et al. (2011) suggested similar mechanism of moss facilitation in conifer regeneration. Moss facilitation could also be argued in light of lack of physical barrier from *P. juniperinum*. The growth form of this and other acrocarpous mosses is erect consisting of single stems, which do not form continuous mat unlike pleurocarpous mosses. The dense stems of acrocarpous mosses have belowground rhizines that binds seedbed substrates together and thereby increase bulk density, which help moisture retention (Brodo et al., 2001). High moisture retention due to high bulk density and cooler surface temperature due to their high shoot density creates favourable seedbed condition for germination and seedling establishment. Bloom (2001) reported higher soil respiration in *Kalmia*-moss community than *Kalmia*-lichen community, which means greater OM decomposition under moss (Sedia and Ehrenfeld, 2006). Higher moisture availability under moss may also facilitate higher microbial activity and thereby increase substrate bulk density.

Although spruce germination was similar in moss intact and moss mixed plots, seedling establishment was higher in moss intact plots. Although seedbed moisture was almost similar between moss intact and moss mixed plots, seedling establishment was lower in moss mixed plots. This can be explained by comparing juvenile seedling mortality between the two treatments due to predation and frost heaving (Allen, 1929; Mallik and Kravchenko, 2016). In this study many seedlings were found partly or wholly uprooted, that were eventually dead at third

count in the first growing season (August 24, 2014) in moss mixed plots despite moist seedbed. Increased seedling predation may occur in moist seedbeds (Nystrand and Granström, 1997, 1998). In dry weather, *Cladonia* lichen can be very dry, from occasional morning dew they can absorb water and expand and germinating cotyledons are often caught by *Cladonia* thalli. Contraction of lichen thalli due to drying may be sufficient to pull some seedlings from the ground. We noticed such seedling uprooting one week after of germination and it was mostly in mat-mixed plots of both lichen and moss dominated seedbed. Ants (*Formica fusca* L.) and curculinoid beetles have been suggested as potential juvenile seedling predators in this area (Power, 2005; Moss and Hermanutz, 2009). Nystrand and Granström (2000) identified two species of carabid beetles consuming Scot pine seedlings in a field experiment in Swedish boreal forest. Wheeler et al. (2011) reported that feather moss protect spruce seedlings from predators during the first two to three years of recruitment.

Our moisture and temperature data showed that moisture content was higher in moss intact seedbed than mat mixed and removed seedbeds which might be responsible for higher germination and seedling establishment in moss intact seedbeds.

4.4. Lichen inhibition

The *in situ* field experiment revealed that lichen dominated seedbed had lower germination and seedling establishment than moss dominated seedbed. The potential mechanism for this might be seedbed moisture, temperature, and chemical effects. Lichen seedbeds had lower moisture but higher surface temperature than moss seedbeds. Furthermore, diurnal temperature fluctuation was higher in lichen than moss seedbeds. Zamfir (2000) showed that, *Filipendula ulmaria* and *Arenaria montana* had lower seedling emergence in both thick and thin lichen mat than on bare soil in dry treatment, but in moist treatment their emergence on thin lichen mat was not different from bare soil. In relatively dry environment, even a thin lichen cover may inhibit seedling emergence of conifers (Splawinski et al., 2018). Fauria et al. (2008) also reported higher temperature on lichen removed soil surface which helped Scots pine (*Pinus sylvestris* L.) growth during early spring, however, their study did not consider seed germination. Similarly, Hébert et al. (2006) reported that, soil water availability in lichen woodland was the limiting factor in slow growth of planted black spruce and Jack pine (*Pinus banksiana* L.) seedlings.

The dark color of OM adsorbs more heat from the sun causing higher surface temperature (maximum temperature reached 40 °C at noon) than moss seedbed (Fig. 8). This high surface temperature in lichen seedbed also contributed to lower moisture content by rapid drying of seedbed. Another possible reason for low moisture in lichen seedbed might be its low bulk density. The OM under lichen mats was composed of unburned lichens and forest floor litter that are spongy, porous and devoid of soil. Unlike forest floor, this unburned and partially burned OM persists for a long time in the seedbed. Sedia and Ehrenfeld (2005, 2006) showed that lichen and plant tissues under lichen decompose more slowly than moss tissue. Furthermore, lichens covering the OM surface can make it water repellent (Brodo et al., 2001).

4.5. Lichen allelopathy

Our laboratory bioassay revealed that black spruce germination, seedling root and shoot length were significantly lower in lichen *C. cristatella* than in *C. stellaris* and the two mosses, *P. juniperinum* and *P. screberi* indicating potential allelopathic effects of *C. cristatella* on spruce seed germination and early seedling growth. This result also resembles that of Pyatt (1967), who suggested that adverse effect of ground lichen, *Peltigera canina* (L.) Willd. on grasses was based on a direct effect of lichen secondary metabolites on root formation and elongation. Gardner and Mueller (1981) reported toxic effects of

Cladonia foliacea on bryophytes. Since lichen allelochemicals negatively affect bryophytes, they might also affect vascular plants. Several *in vitro* studies suggest that the competitive relationships between terricolous lichens and vascular plants depend on allelopathic interference (Lawrey, 1986). In this study, we found no difference in germination, root and shoot length of spruce seedlings among *Cladonia stellaris*, *Polytrichum juniperinum*, *Pleurozium schiberi* and control suggesting the absence of allelopathy in two mosses and the lichen *Cladonia stellaris*. Kytöviita and Stark (2009) also found no allelopathic effect from *C. stellaris* on pine seedlings. Castells et al. (2005) found no allelopathic inhibition of mosses *Sphagnum* spp. and *Hylocomium splendens* on white spruce (*Picea glauca* (Moench) Voss) germination and root-shoot growth. In this study, total phenol content of *C. cristatella* was the highest among the four tested cryptogams. The pH of *C. cristatella* treated medium was lower as well compared to *C. stellaris*, *Polytrichum juniperinum*, *Pleurozium schiberi*. Since germination, seedling root and shoot length were lower only in *C. cristatella* lichen, it is possible that the presence of usnic acid found in this lichen might have inhibited germination and seedling growth of black spruce.

5. Conclusions

This research demonstrates that cryptogamic mats act as a physical barrier to seed germination and seedling growth in older burns (TNR) but not in recent burns (RP and SR). It appears that post-fire moss dominated seedbeds facilitate black spruce regeneration by improving moisture retention and maintaining low surface temperature, while lichen mats inhibit seedling regeneration by creating physically and chemically inhospitable seedbeds. While thick OM and undisturbed micro-sites of oldest site (TNR) helps good seedling regeneration in lichen-dominated seedbeds, thin OM makes comparatively better

seedbed condition in RP and SR sites. The difference between lichen and moss mats with respect to spruce seed germination and seedling establishment is best explained by the altered seedbed microenvironment such as moisture retention and temperature during seed germination and seedling establishment phase. Based on the result from TNR site, one can argue that the degree of lichen inhibition increases time since fire with increased lichen mat density, OM thickness and accumulation of total phenol. This study also demonstrates the possible chemical inhibition of allelochemical, usnic acid on black spruce germination. On the other hand, moss dominated seedbeds facilitate spruce seedling regeneration with increased moisture retention and temperature moderation. Mechanical mixing of lichen mats by eliminating the physical and chemical barrier may create a better seedbed for black spruce regeneration in late post fire sites. However, mat mixing may enhance seedling damage due to frost heaving and animal browsing.

6. Conflict of interest

The authors declare no conflict of interest.

Acknowledgements

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Appendix A

See Figs. A.1 and A.2, Tables A.1–A.3.

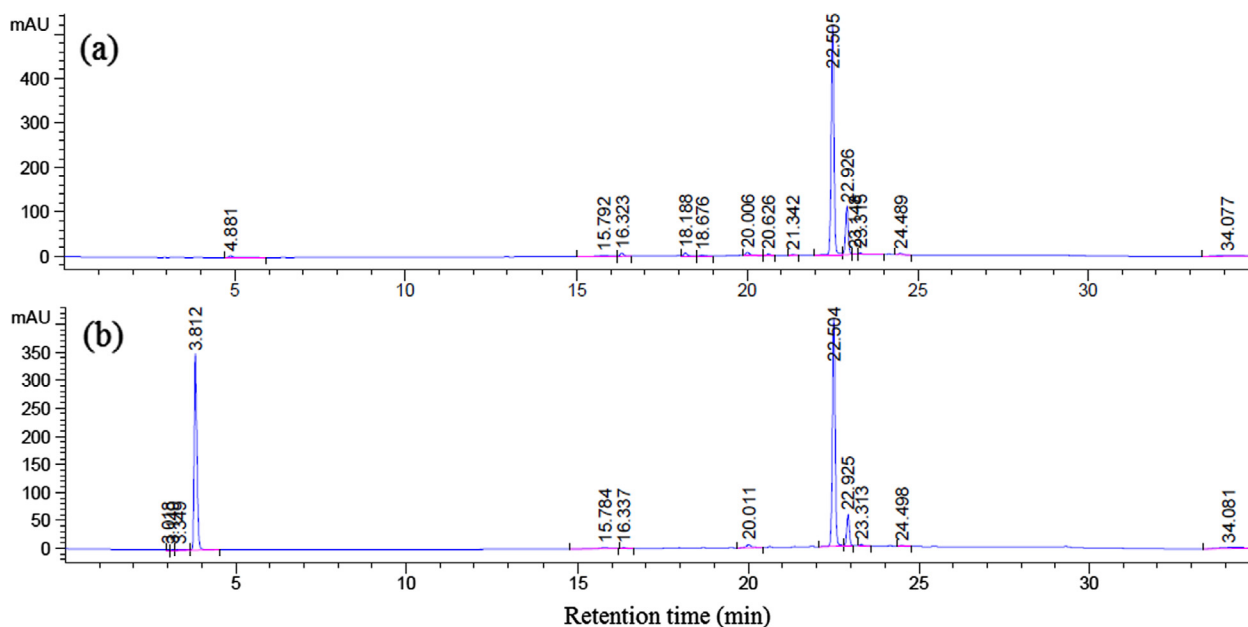


Fig. A.1. Daily rainfall patterns in July and August 2014 in Terra Nova National Park, Newfoundland.

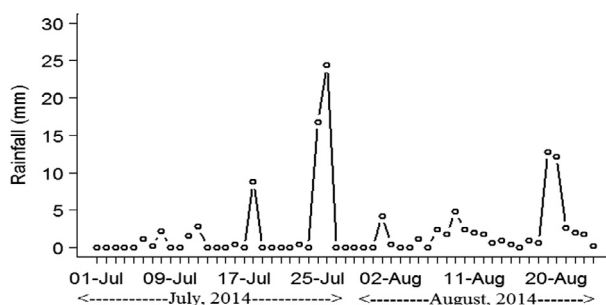


Fig. A.2. Representative chromatograms of *Cladina stellaris* methanol extraction (a) and acetone-dichloromethane extraction (b) at 280 nm wavelength.

Table A.1

Results of repeated-measures analysis of variance showing difference of number of seedling survived by seedbed type, OM thickness and seedbed manipulation treatments among four counts (contrasts) in Spracklin Road, TNNP.

Response		Df	Mean Square	F-value	Pr (> F)
Number of seedlings	<i>Between subjects</i>				
	Seedbed	1	1680.3	15.3	< 0.01
	OM	1	7.5	0.07	0.79
	Treatment	2	2449.8	22.25	< 0.01
	Seedbed × OM	1	2324.1	21.12	< 0.01
	Seedbed × Treatment	2	1086.8	9.87	< 0.01
	OM × Treatment	2	643.5	5.84	< 0.01
	Seedbed × OM × Treatment	2	492.6	4.47	0.01
	Error-1	36	110.1		
	<i>Within subjects</i>				
	Contrasts	3	2082.5	96.47	< 0.01
	Seedbed × Contrasts	3	844.6	39.12	< 0.01
	OM × Contrasts	3	96.4	4.46	< 0.01
	Treatment × Contrasts	6	179.4	8.31	< 0.01
	Seedbed × OM × Contrasts	3	447.7	20.74	< 0.01
	Seedbed × Treatment × Contrasts	6	197.8	9.17	< 0.01
	OM × Treatment × Contrasts	6	82.6	3.82	< 0.01
	Seedbed × OM × Treatment × Contrasts	6	40.8	1.88	0.08
	Error-2	108	21.6		

Significant differences are marked in bold.

Table A.2

Results of repeated-measures ANOVA showing differences in moisture and temperature by seedbed type and substrate depth. Significant differences are marked in bold.

Response		df	Mean Square	F-value	p-value
(a) Moisture	<i>Between subjects</i>				
	Seedbed type	1	3713.0	8.1	0.04
	Substrate depth	1	7443.0	16.1	0.01
	Seedbed type × Substrate depth	1	2107.0	4.6	0.09
	Error-1	4	461.0		
	<i>Within Subjects</i>				
	Date	30	44.4	24.1	< 0.01
	Seedbed type × Date	30	4.8	2.62	< 0.01
	Substrate depth × Date	30	18.5	10.0	< 0.01
	Seedbed type × Substrate depth	30	4.6	2.6	< 0.01
	depth × Date				
	Error-2	120	1.8		
(b) Temperature	<i>Between subjects</i>				
	Seedbed type	1	1745.5	46.4	< 0.01
	Substrate depth	1	84.8	2.3	0.21
	Seedbed type × Substrate depth	1	60.7	1.6	0.27
	Error-1	4	37.6		
	<i>Within subjects</i>				
	Time	47	94.2	17.5	< 0.01
	Seedbed type × Time	47	34.7	6.6	< 0.01
	Substrate depth × Time	47	24.5	4.5	< 0.01
	Seedbed type × Substrate depth	47	2.3	0.4	0.99
	depth × Time				
	Error-2	188	5.3		

Table A.3

Peak summary of HPLC chromatograms of usnic acid standard and lichen *Cladonia cristatella* and *C. stellaris* (RetTime = Retention Time).

Species	Methanol extraction						Acetone-dichloromethane extraction					
	Peak #	RetTime [min]	Width [min]	Area mAU*s	Height mAU	Area %	Peak #	RetTime [min]	Width [min]	Area mAU*s	Height mAU	Area %
<i>Cladonia cristatella</i>	12	22.50	0.09	1844.14	307.4	42.98	3	3.38	0.08	1793.71	323.13	32.96
	14	23.08	0.09	1392.11	233.1	32.45	10	22.51	0.09	1311.28	212.60	24.09
	–	–	–	–	–	–	12	23.09	0.09	1144.68	187.92	21.03
<i>Cladina stellaris</i>	9	22.51	0.08	2902.61	517.1	72.07	4	3.81	0.08	1958.65	351.29	40.22
	10	22.93	0.08	599.48	106.70	14.88	8	22.50	0.08	2274.19	406.93	46.70
Usnic acid standard	7	22.391	0.09	1324.76	221.73	82.85						

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